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AI-Driven Top-Down Modelling: A Pioneering Approach to Fast-Track Development Planning of a Giant Mature Field

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Abstract

GUPCO in collaboration with Dragon Oil, is spearheading a landmark transformation in mature field management through the deployment of advanced Artificial Intelligence (AI). For over five decades, GUPCO has accumulated one of the most comprehensive subsurface datasets in the region, spanning hundreds of wells and extensive production and injection history. Yet, much of this invaluable asset has remained untapped due to the limitations of traditional modeling approaches. This pioneering initiative introduces AI-Driven T.D.M. as a disruptive, high-impact methodology to unlock the hidden value of this data. The goal is to create a fully data-driven, fieldwide predictive model capable of delivering actionable insights, ranging from optimal off-take strategies to high-confidence drilling targets and workover opportunities. Besides, this effort is catalyzing a cultural and technical shift, embedding AI expertise deep within GUPCO's organizational fabric.

At the core of this initiative lies a purely data-driven reservoir modelling framework, free from interpretative biases, geological assumptions, or time-consuming numerical simulations. Leveraging Rosenxt-Intelligent Solutions Inc.'s proprietary TDM technology, the model is trained using empirical field data and refined through advanced AI algorithms including fuzzy logic, artificial neural networks, and proprietary Artificial Intelligent Data Patching (AIDP).

The model undergoes rigorous blind validation using unseen temporal and spatial data to ensure its reliability. Once validated, the model is deployed across hundreds of scenarios to forecast field performance, guide scenario planning, and identify sweet spots for infill drilling. The process is iterative and adaptive, continuously improving with each new data cycle.

Though still in its early stages, the project has already delivered results that are nothing short of exceptional. The AI model has achieved full-field history matching of oil, gas, water rates, and wellhead and reservoir pressures with astonishing accuracy, fully automated and without manual tuning. It has

successfully forecasted future performance through blind testing, demonstrating its strength as a predictive engine.

Hundreds of simulations have been executed, quantifying reservoir uncertainties and mapping optimal development scenarios. More importantly, the model has pinpointed specific high-impact drilling and re-completion candidates, empowering the asset team to act decisively with confidence backed by data.

This study introduces a pioneering AI-based technological advancement to GUPCO, fundamentally transforming the efficiency and depth of technical analyses that underpin field management strategies. This advancement will significantly enhance decision-making capabilities, enabling more rapid and informed responses to operational challenges. Additionally, this initiative represents a strategic investment in human capital, fostering expertise within GUPCO in the application of AI to oilfield operations. As a secondary benefit, the project entails a comprehensive review of GUPCO's extensive data archive, rendering it a more structured and actionable resource to support future initiatives and assessments. Ultimately, this initiative positions GUPCO at the forefront of AI-driven oilfield management in Egypt, setting a precedent for technological innovation within the industry.

Introduction

Reservoir Description

The subject field is a giant mature oil field discovered in 1965 in the central part of Gulf of Suez (GoS) in Egypt. The 'K' reservoir and the overlying 'B' reservoir are the main reservoirs in the field. The upper reservoir, 'B,' was deposited as a fan-delta and consists of set of coarsening upward sand deposits with lamination of silts and dolomites. Four sub-units make up the 'B' reservoir, however, most of production is attributed to the middle part of the top sub-unit. The underlying 'K' reservoir was deposited as a huge wave-dominated fan delta and consists of five main sub-units. The 'K' reservoir hosts most of the oil in place and has generally good quality: overall net-to-gross ratio averaging 50% and reaches 90% in the best parts of the reservoir; average porosity is ~22%; permeability up to few Darcies. Type log is shown in [Figure 1](#).

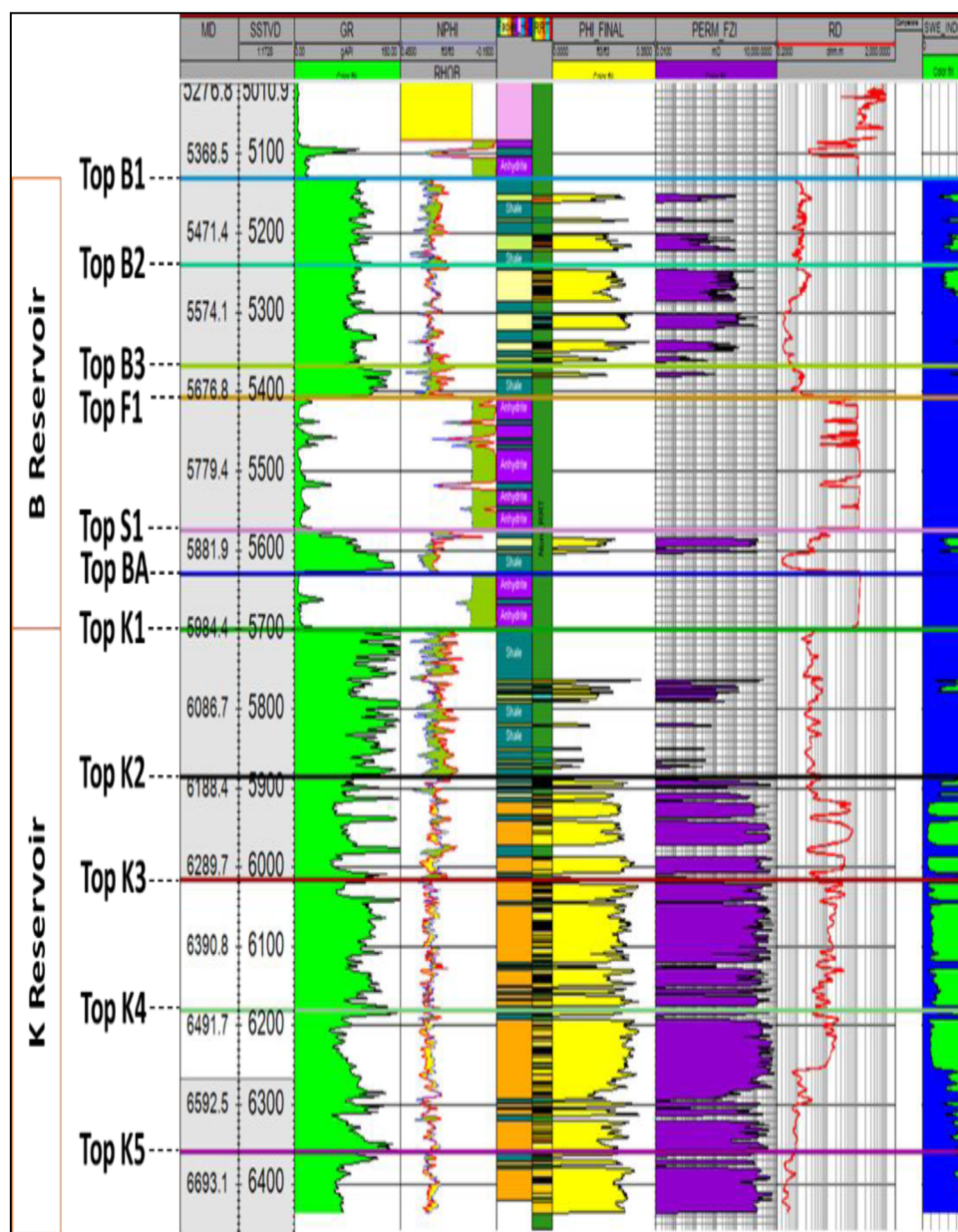


Figure 1—Type Log

The structure map of the reservoir is shown in Figures 2. The mapping heavily depends on well density to complement the poor ability of seismic to image the reservoir due to presence of shallower salt formations.

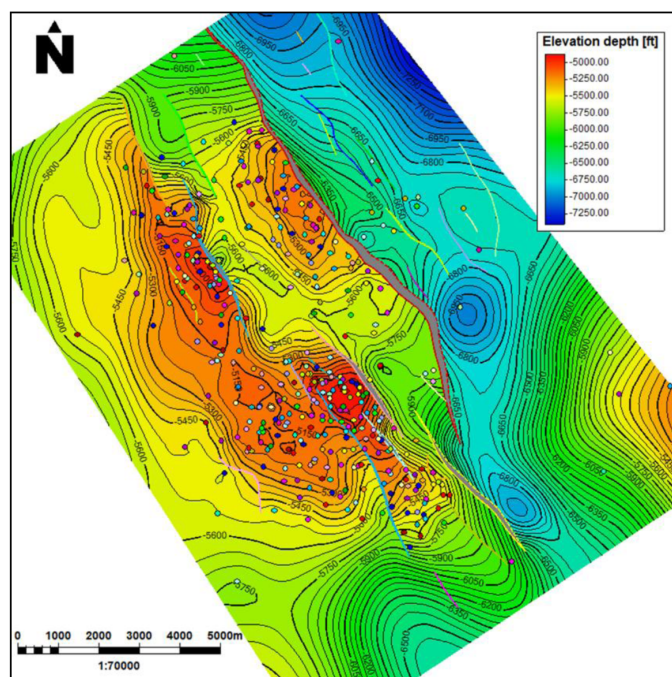


Figure 2—Top Structure Map of Reservoir for South Part of the Field

Field Performance

The subject field had a strong production start-up in late 60's, reaching peak oil production of ~320,000 bbl/d, however reservoir pressure dropped quickly, hence field was temporarily shut down and water flooding was initiated to support reservoir pressure. Figure 3 shows field production/injection history.

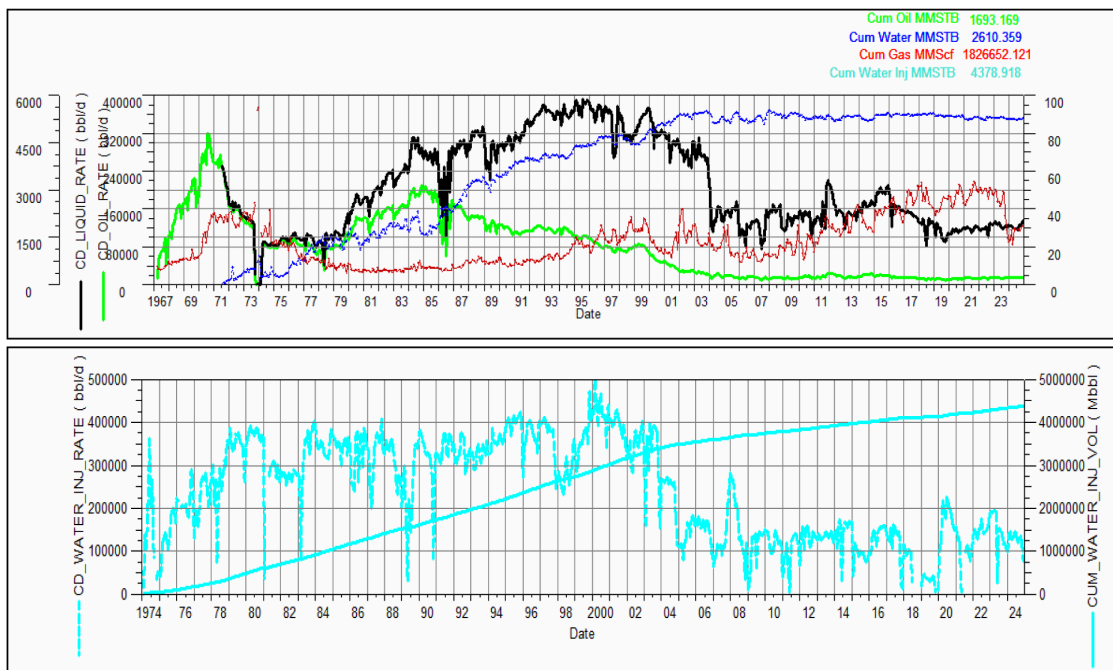


Figure 3—History of Production and Injection

Historically, GUPCO has no dynamic numerical reservoir simulation models (DNRSM) for the subject field, and it will take ages and lots of investment to build and calibrate such models that most probably be

based on plenty of uncertain assumptions. By the time GUPCO manages to build decent DNRSRM, Dragon Oil (DO) is perhaps getting close to the end of the concession period.

GUPCO collected huge amounts of very valuable data during 58 years of operating the field, which are ideal for developing big data driven AI technology-based models. The appropriate integration of AI in the subsurface evaluation of expansive hydrocarbon fields, such as in GOS, is increasingly acknowledged for providing robust depictions of current and projected future performances. The AI technology can produce accurate models in a short turnaround time and its physics-based deep learning capability of this technology can accurately gauge the importance of collected data (away from any human assumptions)

Project Description

Background

The ultimate objective of this project is to develop a history-matched reservoir simulation model for the M and B Fields using Rosenxt-ISI's IImagine Software's technology. More specifically, the objective of this project is to build reservoir models for the M and B Field that would use the existing information from the fields in order to predict the production and field performance throughout the reservoir as a function of reservoir characteristics and operational constraints that are employed in the field.

This project includes a six-tasked plan to accomplish the objectives set forth by Dragon Oil Company as below:

1. "Task 1" Data Collection, Transfer, and Quality Checking
2. "Task 2" Pre-Processing, Quality Control and Data Preparation
3. "Task 3" AI-based (Top Down) Reservoir Simulation and Modeling
4. "Task 4" AI-based Production Allocation
5. "Task 5" Oil Production Optimization
6. "Task 6" Final Report of the Entire Project

This paper provides the details of Task 3 of the M and B Field Projects. First, a brief theoretical discussion is provided about Top-Down Modeling. The AI models were designed based on the data availability of the analyses performed in previous tasks. With the model designed, the data quality and quantity are used to determine the topology and architecture of the machine learning algorithms for both wells in the B and K Reservoirs

Upon completion of the above tasks, training and calibration of both the B and K AI models were initiated. During this process, the production and injection data along with operational constraints and static reservoir characteristics were used to history-match production from each well in each reservoir. Through the training of this task, the team has identified the most appropriate set of input variables that can result in the best history-match and calibration of the AI model. Once this selection was made and satisfactory training and calibration of the AI models were completed, the models were validated using the tail end of the production from the active wells as a blind history-match. The results of a selection of wells for B and K are shown within the paper.

Theoretical Discussion about TOP-DOWN Modelling

Top-Down, Intelligent Reservoir Modeling or Top-Down Modeling (TDM) is a novel reservoir simulation and modeling technology that was developed and recently introduced to the exploration and production industry. Since its introduction, the results of Top-Down Modeling have been peer-reviewed and published in multiple occasions. TDM may be defined as formal, comprehensive empirical modeling of fluid flow in the hydrocarbon reservoirs.

Imagine a team of reservoir engineers, reservoir modelers and reservoir managers with a significant amount of collective experience working on a brown field from day one. After many years of production and injection in this field, under their supervision, they have developed a relatively deep understanding of the way this particular field behaves. Members of this team would usually be able to see the production/injection behaviors in individual wells (no matter how strange it may seem to others, at times) and make sense of it and be able to interpret these behaviors. Of course, this team of experts is usually armed with a wealth of geological information, well logs, cores analyses, well tests, and production and injection history. One of the major shortcomings in such cases is usually the fact that the expertise that we are talking about is located with these experts and is usually applied on a localized basis (one or at best a few wells at a time).

Now Imagine being able to capture the essence of all the expertise we talked about in the previous paragraph in a comprehensive and cohesive manner within a formalized reservoir modeling framework that can be trained, calibrated and validated to match the production history of the field and forecast its future behavior. Top-Down Modeling is such a technology. Top-Down Modeling is a formalized reservoir simulation and modeling technology that learns from real, historical data instead of mathematical modeling of our current understanding of the underlying physics of reservoir behavior.

The objective of Top-Down, Intelligent Reservoir Modeling (henceforth Top-Down Modeling) is to provide operators with an independent model of the fluid flow in the reservoir. The Top-Down Model should be used in concert with the existing numerical models and other tools at the engineers' disposal in order to help the asset team in making informed decisions regarding optimum operation of the assets.

In order to put the concept of Top-Down Modeling in reservoir engineering and reservoir modeling context, let us define production as a function of reservoir and fluid characteristics, operational constraints and other variables in the following formulation:

$$q = f(x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n, w_1, w_2, \dots, w_n) \quad \text{Equation 1}$$

Where

q = production from the reservoir

$x_1, x_2, \dots, x_n$ = Reservoir & Fluid Characteristics

$y_1, y_2, \dots, y_n$ = Operational Constraints

$w_1, w_2, \dots, w_n$ = Other variables

$f(x)$ = Constant and predetermined functional relationship

The above equation simply states that from a reservoir engineering/modeling point of view, production from a field is modeled by using a functional relationship between reservoir and fluid characteristics (formation thickness, porosity, permeability, viscosity, compressibility, etc.), operational constraints (drilling new wells, injecting water, shutting some wells, changing the surface facility capacity, etc.) and some other variables such as well configurations, different completion techniques, etc.

	Numerical Model	Top-Down Modeling
Reservoir Characteristics	Uncertain: • Measurements • Interpretations (subject to modification during the history matching)	Uncertain: • Measurements • Interpretations (subject to modification during the history matching)
Functional Relationships	Certain: • Conservation of Mass • Darcy's Law (unchanged during the history matching)	Uncertain: • Relationship between reservoir characteristics and production. (subject to modification during history matching)

Figure 4—Difference between assumptions made in numerical modeling and Top-Down Modeling.

This formulation holds and is common for both numerical reservoir simulation and Top-Down Modeling. In both of these modeling technologies we try to model production as a function of reservoir-fluid characteristics, well characteristics and operational constraints. The major difference between these two technologies is in the assumptions that are made regarding the state of our knowledge of the fluid flow in porous media phenomenon. This difference in assumptions is demonstrated in Figure 4.

Conventional reservoir simulation and modeling is a bottom-up approach. It starts with building a geological model of the reservoir that is populated with the best available petrophysical and geophysical information at the time of development. Engineering fluid flow principles are then added and solved numerically so as to arrive at a dynamic reservoir model. The dynamic reservoir model is calibrated using the production history of multiple wells and the history-matched model is used to strategize field development in order to improve recovery.

In the conventional numerical simulation and modeling the functional relationship used in Equation 1 consists of the law of conservation of mass, Darcy's law (Fick's law of diffusion in the cases that such formulation is required), thermodynamics and conservation of energy (if we are modeling thermal recovery), etc. These functional relationships are believed to be true, deterministic and unchangeable. Therefore, if the production rates (as a function of time – production history) that results from this formulation does not match our observation from the field (measured rates), we conclude that the reservoir characteristics may not be ideally measured and interpreted, and they must be modified in order to achieve a match.

We are not disputing the validity and application of this technology but would like to point out that the functional formulation that is coded into our numerical reservoir models has evolved from a simple relationship in the early days of reservoir simulation (single-phase, Darcy's law) to a much more complex set of relationships that models more complexities in the reservoir (multi-phase flow, dual porosity formulation, stress-dependent permeability, compositional formulation, coupling with geo-mechanics and surface facilities, etc.) and it is bound to evolve even further as our knowledge of these physical phenomena deepens. Good evidence for this claim is newer and better versions of the reservoir simulation software packages that are released routinely.

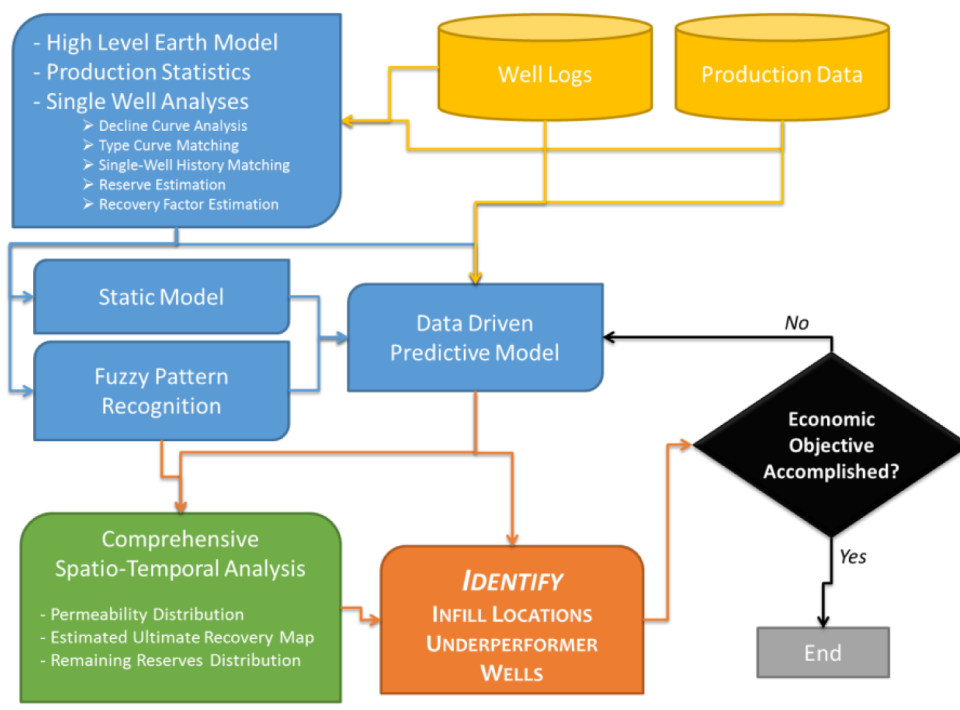
Therefore, during the history-matching of a numerical reservoir simulation model, since the functional relationships are constant and unchangeable (i.e. our understanding of the physical phenomena is good enough that does not need modification no matter which reservoir we are modeling) the engineer concentrates on modification of reservoir characterization (such as permeability) in order to reach a reasonable match. Since the reservoir characterization is represented by a geo-cellular model developed by geoscientists and is full of interpretations and "fuzzy" values, we as engineers feel comfortable changing these numbers in order to get the match we are looking for. Please note that we are not criticizing this approach but merely stating the process in order to emphasize its differences with Top-Down Modeling.

Top-Down Modeling, a newly introduced and peer-reviewed reservoir modeling technology, approaches the reservoir simulation and modeling from an opposite angle by attempting to build a realization of the reservoir starting with well production behavior (history). The production history is augmented by core, log, well-test and seismic data in order to increase the accuracy of the Top-Down Model. Although not intended as a substitute for conventional reservoir simulation, this innovative and novel approach to reservoir modeling can be used as an alternative to conventional reservoir simulation and modeling. In cases where a conventional model of a reservoir already exists, Top-Down Modeling should be considered as a compliment to, rather than a competition for, the conventional technique, to provide an independent look at the data coming from the reservoir/wells for optimum development strategy and recovery enhancement.

In Top-Down Modeling, we modify some of the assumptions that are made in conventional numerical modeling. Instead of holding the functional relationship constant, we allow these relationships to change as well as the possibility to modify the reservoir characteristics. In other words, we refrain from using constant, deterministic, and non-flexible functional relationships between production and reservoir characteristics. In Top-Down Modeling, we try to identify the functional relationship that would produce the type of production

from the reservoir that we have observed (actual field production) using the set of measured reservoir characteristics.

Of course we can, and usually will, modify reservoir characteristics if we believe that one or more sets of reservoir characteristics (measurements) are better than the one we are using. Once we have identified a set of reservoir characteristics that we are reasonably comfortable with (usually using only measured and not interpreted data unless we have high degree of confidence on a particular interpretation) during the history-matching process, we do not change these reservoir characteristics. Instead, we let the functional relationship to be modified until the match is accomplished.



Flow Chart for Top-Down, Intelligent Reservoir Modeling

Figure 5—Top-Down, Intelligent Reservoir Modeling workflow.

The most common criticism to this approach (usually put forward by engineers) is often posed in the form of: "so, what about physics!!!" The average engineer, when confronted with this approach, feels uncomfortable that the basic laws of physics have not been used explicitly during the modeling process. After all, the conventional definition of modeling is the mathematical formulation of a physical phenomenon. The essence of data-driven modeling is the deduction of the physics of the phenomenon from the representative data that is used to construct it. Therefore, physics is honored by data.

The Top-Down Model is calibrated (history-matched) using the most recent set of wells that have been drilled. The calibrated (history-matched) model is then used for field development strategies to improve and enhance hydrocarbon recovery.

The Top-Down Modeling is an elegant integration of the state of the art in Artificial Intelligence & Data Mining (AI&DM) with sound reservoir engineering technology and principles. It provides a unique perspective of the field and the reservoir using actual measurements. It provides qualitatively accurate reservoir characteristics that can play a key role in making important and strategic field development decisions. Figure 5 shows a general flow chart for Top-Down Modeling development.

In summary, a general comparison between conventional numerical reservoir modelling and TDM is shown in Table 1.

Table 1—Comparison of Conventional Numerical Reservoir Modelling and TDM

Numerical Reservoir Modelling	Top Down Modelling (TDM)
Includes simplifications, assumptions and interpretations	Does not make any assumptions and interpretations
Does not use all the Field Measurement Data	Can use any field measurement Data: from reservoir, every well, surface facility, ...
Requires geological model (geological model is modified for history matching)	Does not need geological model (It can generate geological model)
History Matching is not automated (requires manual modification of geological map)	History matching is automated
Development takes months to years	Development usually takes weeks to a few months
Deployment takes hours (on super computers)	Deployment takes minutes (on laptops)

Outlier Identification

As part of the data quality assurance process, this section outlines the steps taken to identify and address outliers within the dataset. Although an initial cleaning was conducted using global minimum and maximum thresholds, further manual inspection was necessary.

This additional review aimed to detect anomalous values that fell outside of expected operational or physical behavior but were not captured by global thresholds. Identifying and resolving these outliers was a critical step to ensure the reliability, accuracy, and consistency of the data used in subsequent modeling and analysis tasks.

Well biographies, developed during previous tasks, served as a primary tool for visually and analytically identifying anomalous data points. These biographies allowed for the detection of inconsistent trends and unrealistic values across key dynamic parameters. Upon validation, any data points confirmed as erroneous were systematically removed from the dataset to maintain data accuracy and minimize the risk of skewed model outcomes. Example is shown in the following figure.

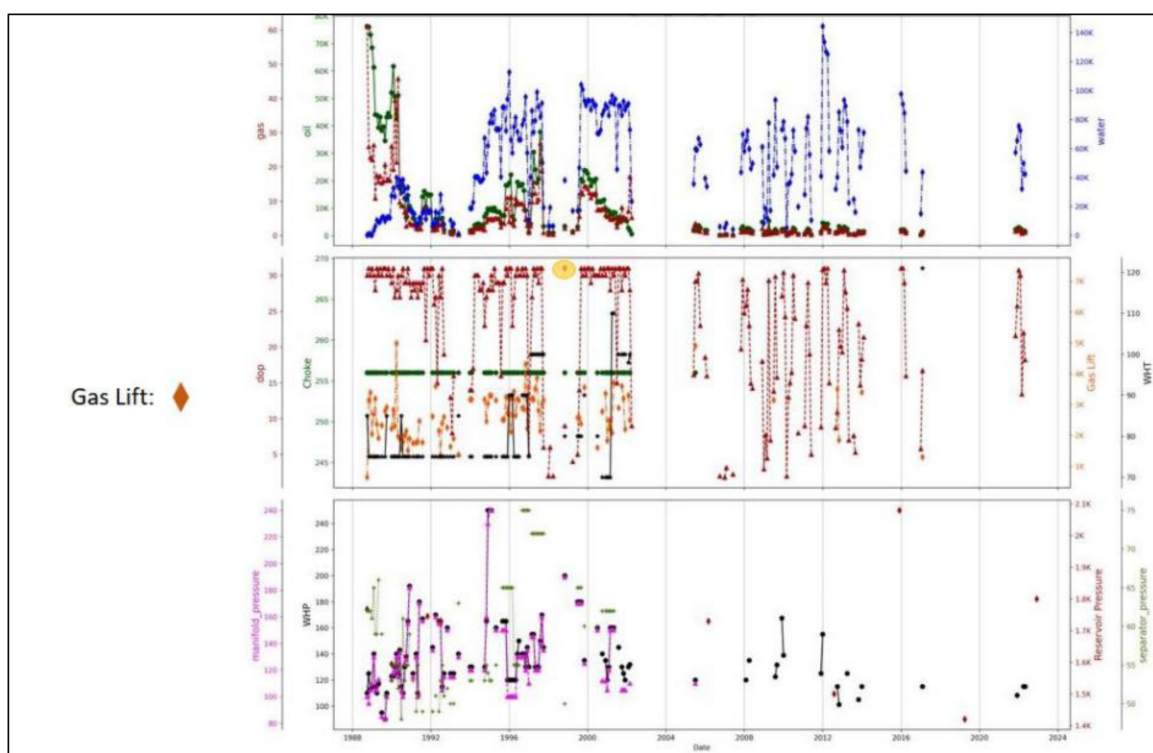


Figure 5—Example of well biography with some records identified as outlier

Artificial Intelligence Data Patching (AIDP)

Following the removal of confirmed outlier records, a comprehensive review of the dataset was conducted to identify variables with missing values. These missing data points originated from two primary sources:

1. Records that were initially absent in the original dataset.
2. Records that were removed during the outlier identification process due to quality concerns.

Figure 6 illustrates the number of available versus missing records for key dynamic variables across both K and B reservoirs. This analysis played a critical role in guiding the selection of variables to be included in the Top-Down Modeling (TDM) process. As a general rule, any dynamic variable with less than 25% data availability was excluded from modeling consideration, except for reservoir pressure, which was retained due to its high importance despite much lower data density.

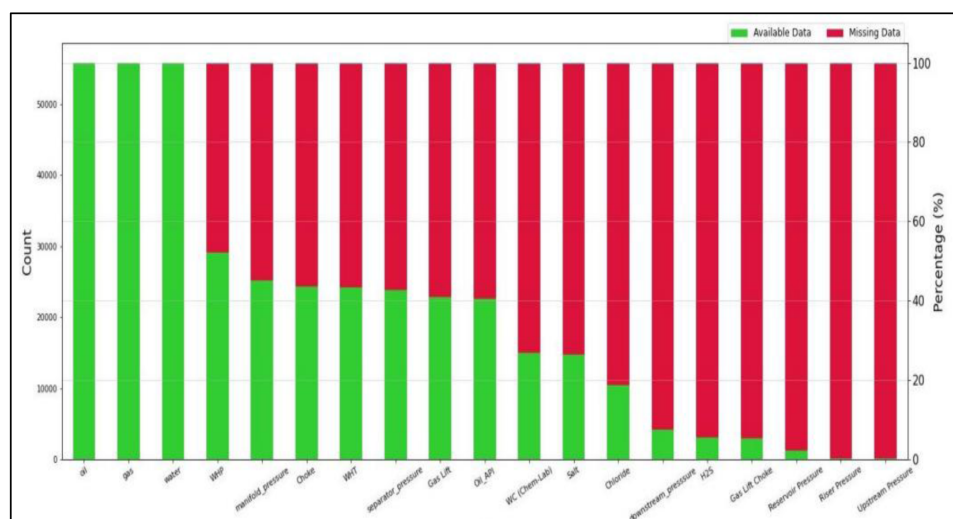


Figure 6—Number of available and missing records for dynamic variables, along with the percentage (K and B)

In addition, Figure 7 presents a breakdown of available and missing records for static variables. This evaluation was essential to ensure the selection of a robust and well-supported input set for the AIDP and TDM processes.

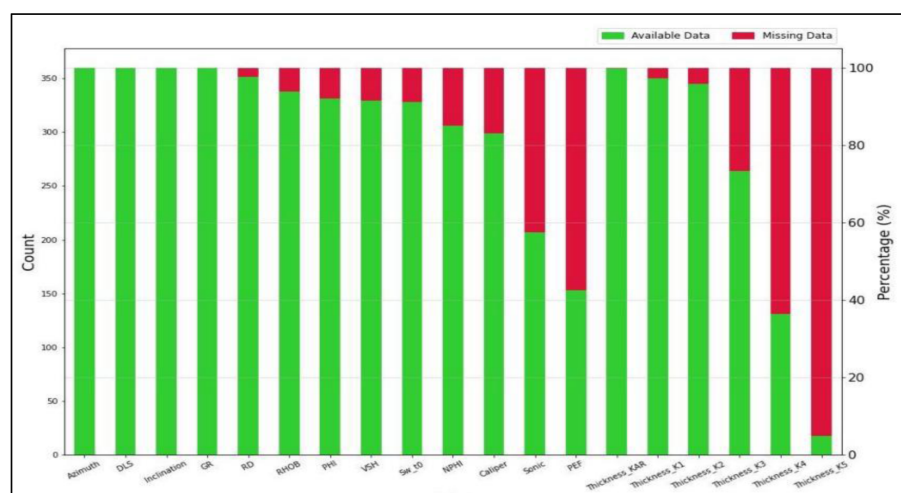


Figure 7—Number of available and missing records for static variables, along with the percentage (K Reservoir)

Petrophysical Variables

Artificial Intelligent Data Patching (AIDP) is a systematic approach that fills in missing variables sequentially, starting with those that have the highest percentage of available data and progressing to those with the lowest. The reason for this approach is that there is a complex relationship between each attribute and this approach provides more confidence in the sequential patching process. Since well logging has been consistently performed throughout the life of the reservoirs, they had the least amount of missing data and were considered as the starting point of AIDP. However, as the frequency of logging tool measurements are different, i.e. GR is measured more frequently than Photoelectric, static AIDP also needs to be patched from the lowest missing toward the highest. The following table shows the availability of petrophysical variables in K and B.

Table 2—Percentage availability of petrophysical (static) data in B and K

Reservoir	GR	RHOB	PHI	VSH	NPHI	Sonic	PEF
B	97%	95%	95%	93%	92%	73%	63%
K	100%	93%	91%	91%	85%	57%	42%

To validate AIDP's reliability, random wells were held out as blind validation wells. These wells were excluded from training, and their petrophysical properties were predicted by the trained model and compared to the actual values to assess the accuracy of the model.

For the training process, the remaining wells after excluding the blind ones are divided to two groups of training and calibration typically with proportion range of 70% to 90% for training and 10% to 30% for calibration. The calibration dataset is used as a criterion to stop the training process to avoid over-fitting (memorization) of the data. [Figure 8](#) shows a sample of Shale Volume (VSH) training and its performance on the trained, calibrated, and blind validated dataset along with feature importance in K. After training, the blind validation well predictions were compared with their actual data and if satisfactory, that model would have been used to patch the missing petrophysical variable for the missing wells. [Figure 9](#) shows the blind validation results of each petrophysical variables.

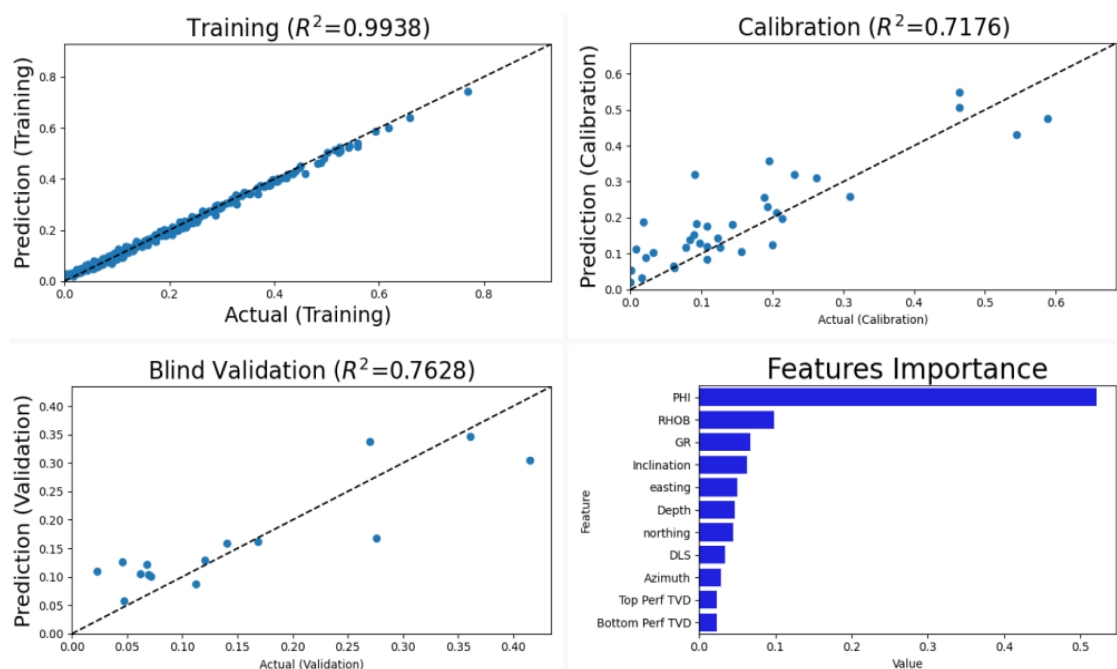


Figure 8—Training results for VSH in K

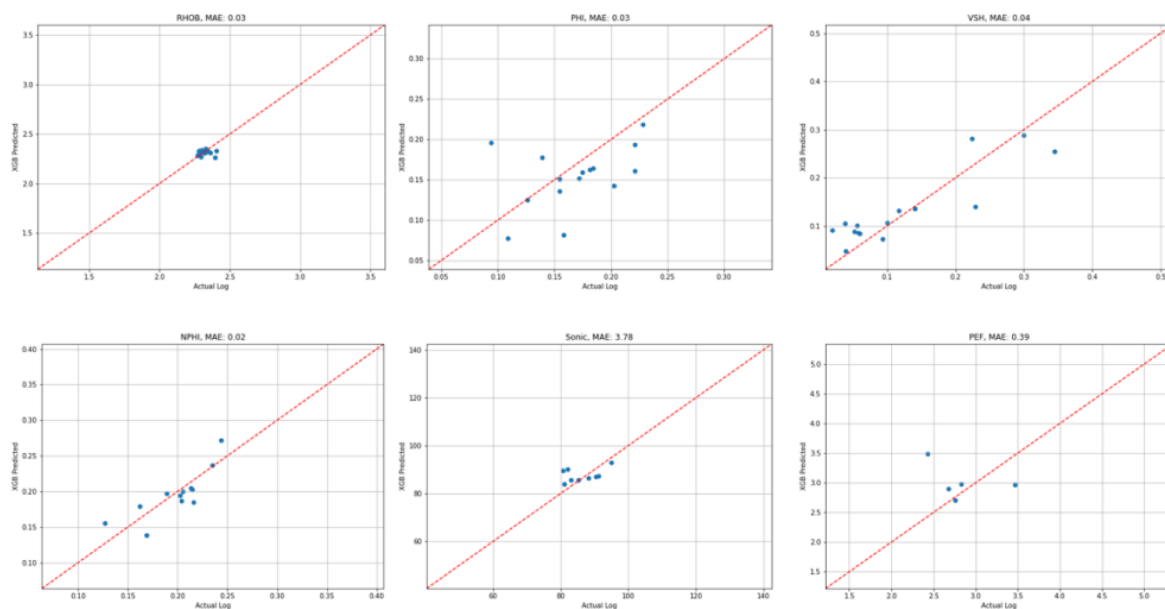


Figure 9—Petrophysical blind validation results for K

Dynamic Operational Data

For Artificial Intelligent Data Patching (AIDP) of dynamic variables, a similar approach to static parameters was taken. First, availability of each variable was identified, then important variables to be patched were chosen, and finally order of patching was identified based on the order of availability. It was decided to patch WHP, WHT, Gas Lift and Reservoir Pressure respectively. Aside from Choke, which is a human-controlled operational variable, the others (WHP, WHT, Gas Lift, and Reservoir Pressure) are patched using AIDP algorithms. Similar steps that were used for AIDP of static variables are used for dynamic ones. Multiple wells were completely held out as blind validation, and the rest of data was partitioned into two categories of training and calibration. Then the model is trained based on the selected input variables, and each category's

performance is evaluated. To be able to trust a model and use it eventually for AIDP, the blind validation performance is the key to rely on. This process is repeated until satisfactory blind validation is obtained.

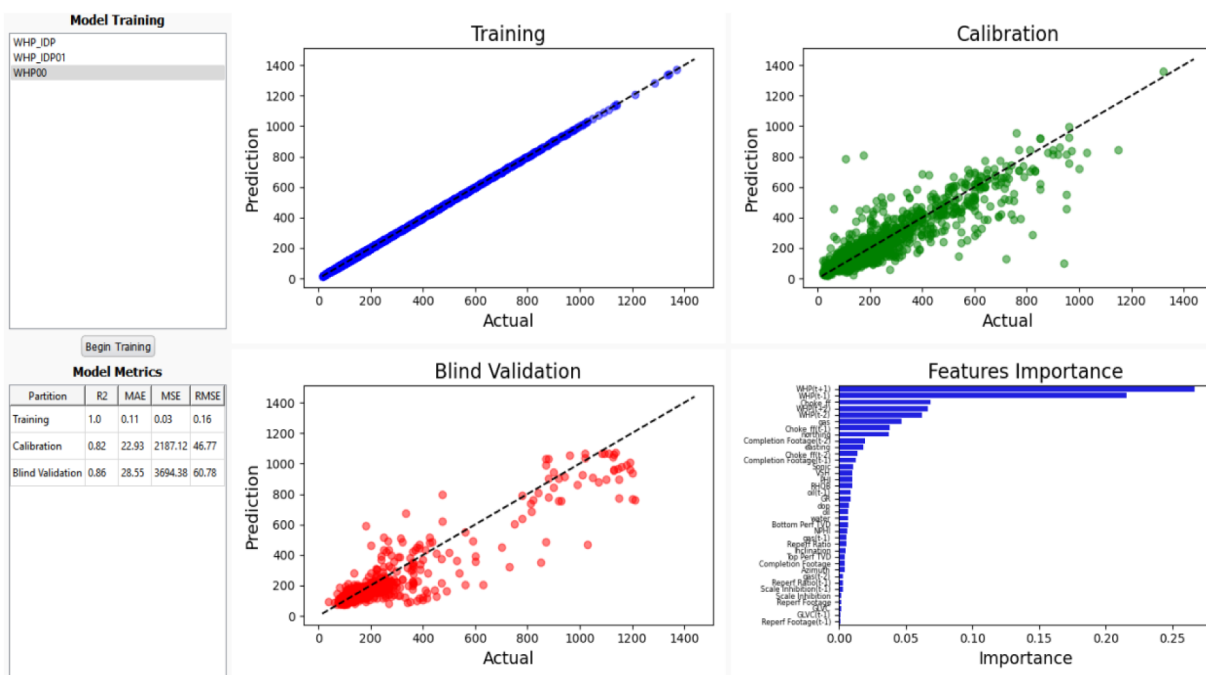


Figure 10—Training result of WHP alongside feature importance in K

Figure 10 shows the final training results of WHP for K as a sample. This figure shows the quality of training on all the datasets of training, calibration, and blind validation wells. Moreover, it shows the ranking (importance) of each feature used as input for the AIDP training process for WHP.

TDM History-Matching Strategy

To identify all the required details of the "Final History-Matched Top-Down Model", a strategy was developed for the history-matching process. This strategy included creating a model using blind validation in time to identify the following:

- The most appropriate set of hyper parameters for the predictive analytics
- The best set of input parameters for the predictive analytics
- The most appropriate number of offset wells for the predictive analytics

In Top-Down Modeling, to develop a robust reservoir model and to identify the necessary parameters for modeling, only a portion of the data is used for history-matching process and a certain amount of data is left out of this process to test the predictive capabilities of the reservoir model. This process (leaving out a certain amount of data) is done to learn about the required TDM parameters for the most robust modeling.

During the development of the TDM, production data was available for more than 50 years. In order to fulfill the objective of the "Blind Validation in Time", the last couple of years were kept out of model development period. This analysis was repeated for 1, 3, and 5 years to establish a foundation on how many years each model is able to forecast reliably before error propagation increases and confidence in the model drops. Below figure shows the timeline of TDM strategy. The rest of the dataset (before blind validation period) is split into training, calibration, and validation partitions randomly.

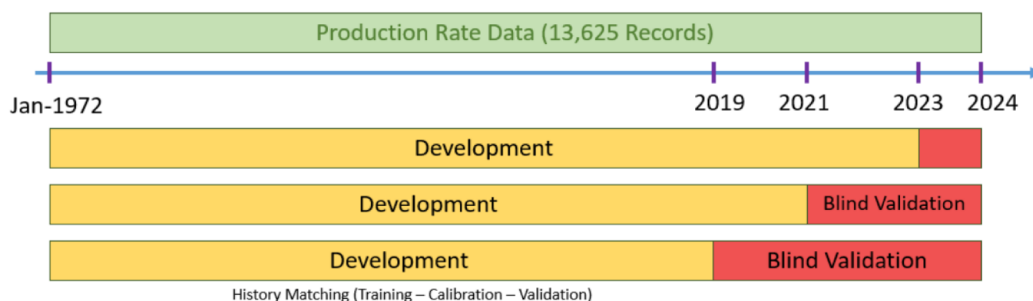


Figure 11—Timeline of different TDM strategies.

Figures 12 through 14 demonstrate the results of the entire field oil history-matching with blind validation in time for 1, 3, and 5-year, respectively. These results suggest that the model provides reliable forecasts up to approximately three years, beyond which the uncertainty increases. While forecasts beyond this period can still offer valuable insights, they should be interpreted with caution, keeping the growing uncertainty in mind.

As TDM uses past (historical) data to forecast, the error can propagate in time. Thus, the question would be how long the forecast can be reliable? The general answer would be the duration of the ‘Blind Validation’. During history matching process, many models can be obtained that have a great match. However, there might be one (if any) that can be considered as a good match for the ‘Blind Validation’ period. Therefore, Blind validation is the yardstick which tells first if a model is reliable and second how long it can be relied on. Figures 12 and 13 show that the model follows the production pattern with a very good accuracy for 1 year and 3 years during blind validation period while it starts deviating after that as shown in Figure 14. This can indicate that the model will be reliable for forecast up to 3 years and after that the confidence in model forecast drops.

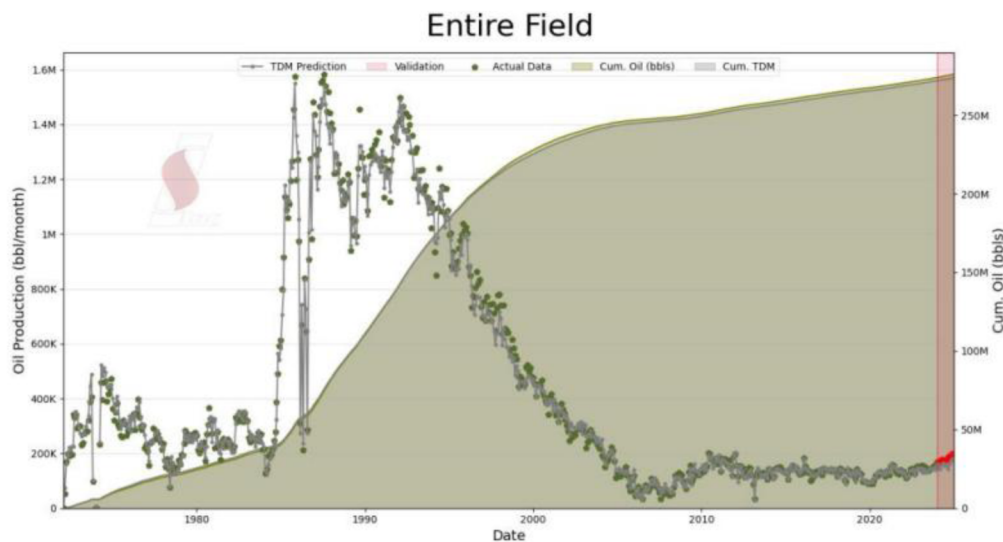


Figure 12—Entire Field history-matching results with 1-year Blind Validation (B)

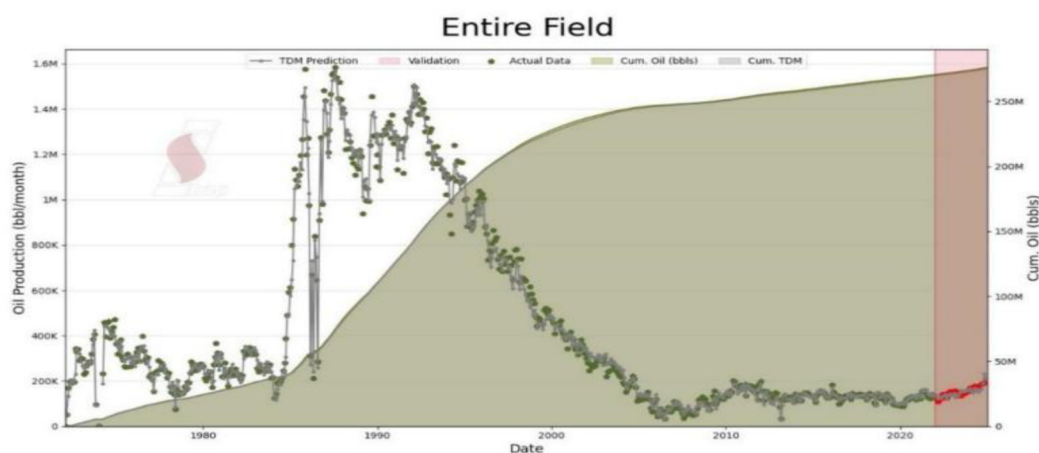


Figure 13—Entire Field history-matching results with 3-year Blind Validation (B)

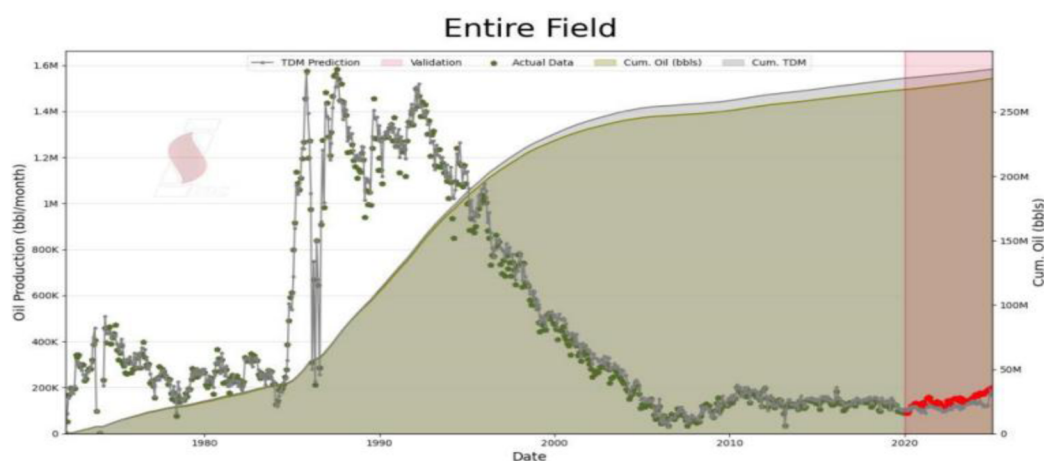


Figure 14—Entire Field history-matching results with 5-year Blind Validation (B)

Top-Down Modelling Design

The blind validation in time, discussed in the previous section, provided critical insights that guided the TDM development team in identifying the optimal configuration of parameters needed to build the final Top-Down Models (TDM). As a result of this analysis, it was determined that a three-year blind validation (BV) period offers the best balance between training and forecasting performance. Additionally, the final model was designed to predict five key dynamic variables: oil rate, gas rate, water rate, reservoir pressure, and wellhead pressure (WHP). Figure 15 shows 5 predictive models that construct the full field reservoir model (TDM).

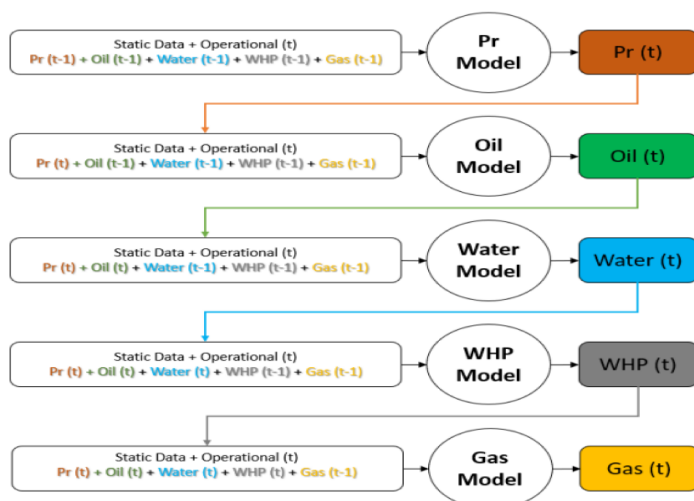


Figure 15—TDM Structure with 5 predictive models sequentially connected to each other

The left chart in Figure 16, and Figure 17 show the entire time of oil, and water rate along with TDM results and cumulative production, and the right chart shows the zoomed version. Figure 18 also demonstrates the location of each well with its polygon. The color of polygon in the left plot shows the well type (producer or water injector).

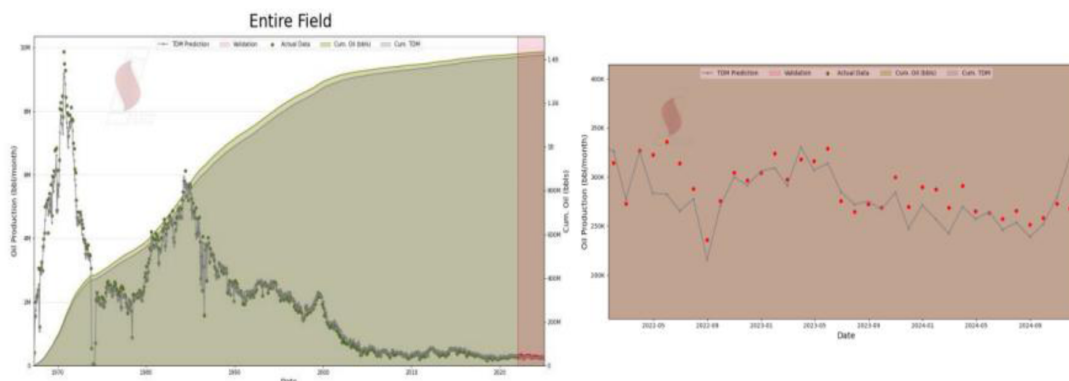


Figure 16—History-matching of oil rate with 3 years of Blind validation for the entire field for K Reservoir

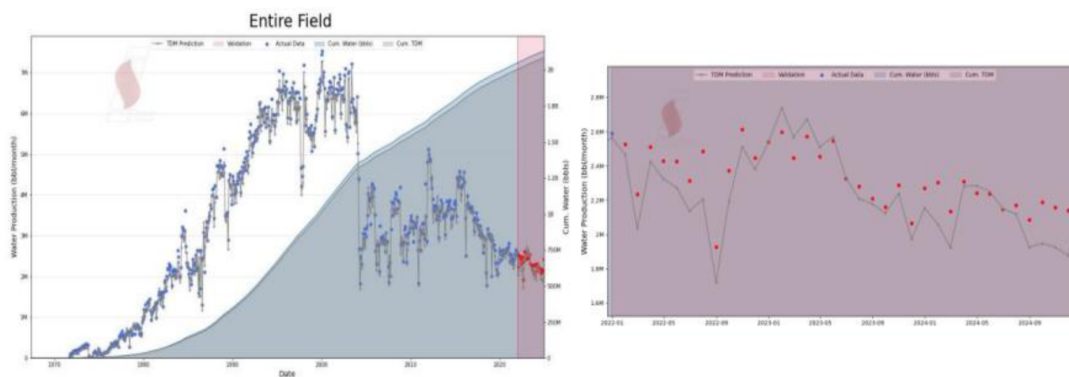


Figure 17—History-matching of water rate with 3 years of Blind validation for the entire field for K Reservoir

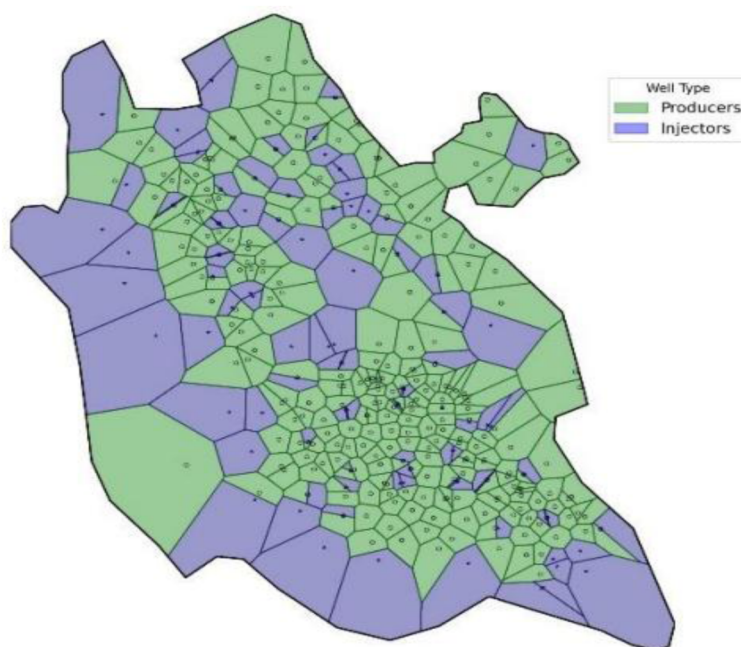


Figure 18—Well Type plot for K Reservoir

Figure 19 shows History-matching of oil, gas, and water rate (top rows), well location, WHP, and reservoir pressure (bottom rows) with 3 years of Blind validation for one of the wells. Results show great history-matching results with acceptable accuracy during the last 3 years of blind validation. This multi-panel visualization helps illustrate not only production trends, but also the well's spatial context and pressure behavior, offering a comprehensive view of the model's performance at the well level. The overall results of all the individual wells clearly demonstrate the power of Top-Down Modeling to history-match complex mature fields with accuracy that are far superior to traditional numerical reservoir simulation.

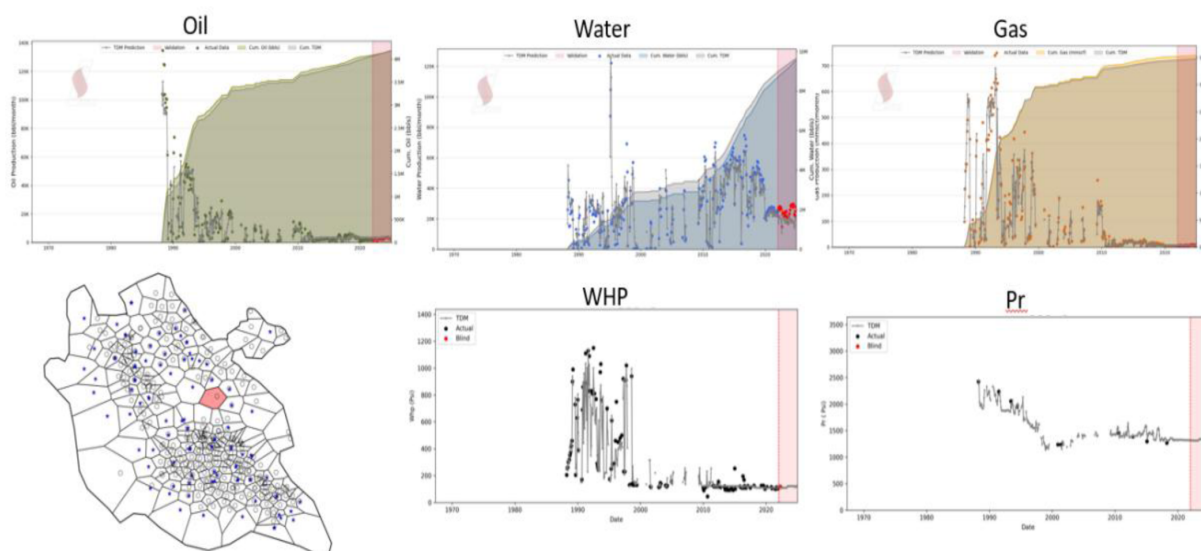


Figure 19—History-matching of oil, gas, and water rate (top rows), well location, WHP, and reservoir pressure (bottom rows) with 3 years of blind validation

Summary/Conclusion

During the "Task 3" of this project, the team developed a comprehensive spatio-temporal database from all the measured data. The spatio-temporal database was uploaded into IImagine™ software application for the development, history-matching, and validation of a series of data-driven reservoir models (also known as Top-Down Models). The development, history-matching, and validation of the Top-Down Models were successful, and the results are presented in this paper. The Top-Down Model simultaneously history-matched five major dynamic variables from the field on a monthly basis. These dynamic variables are: 1. Oil Production Volume 2. Gas Production Volume 3. Water Production Volume 4. Reservoir Pressure 5. Well Head Pressure. The results presented in this paper are of acceptable accuracy and provide a reasonable model for field development planning and production optimization that will be performed during the "Phase Five" of this project.

Way Forward

During "Task 4" of the project, production and injection allocation will be done to identify the contribution of each layer in B (B1, B2, and B3) and K (K1, K2, K3, K4, and K5) reservoirs. This allocation is essential for further analysis and optimization of the field during "Task 5" where field development planning will be performed.

References

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